Human Activity Recognition (HAR) using CNN and Transfer Learning (MobileNetV2)

Abstract:

The primary objective of this project is to develop an automated system for Human Activity Recognition (HAR). Recognizing activities like Running, Walking, and Standing is essential for applications in intelligent surveillance, healthcare, and human-computer interaction. In this work, we implemented a Convolutional Neural Network (CNN) based approach utilizing Transfer Learning with the MobileNetV2 architecture. The model was trained on a dataset of over 2,300 frames extracted from video sequences. To enhance the model's robustness and prevent overfitting, extensive Data Augmentation (rotation, shifting, and flipping) was applied.

The final system achieved a validation accuracy of 81.5%. The model was further tested on unseen video files, demonstrating its ability to accurately classify human movements in real-time environments. The results show that using pre-trained weights from MobileNetV2 significantly reduces training time while maintaining high precision for mobile and edge-computing applications.

3. Methodology & System Design:

3.1 Data Preparation & Preprocessing

The raw data for this project consists of video clips. Since deep learning models cannot directly process video files, a frame extraction technique was used:

- **Frame Extraction:** Using OpenCV, every 10th frame was extracted from the videos to create a diverse image dataset.
- **Image Resizing:** All frames were resized to 128 × 128 pixels to maintain consistency and reduce computational load.
- **Normalization:** Pixel values were scaled from [0, 255] to [0, 1] to help the model converge faster during training.

3.2 Model Architecture

For this project, we used Transfer Learning with MobileNetV2. MobileNetV2 was chosen because it is highly efficient and optimized for mobile and real-time applications.

- **Base Model:** Pre-trained weights from the ImageNet dataset were used to extract high-level features.

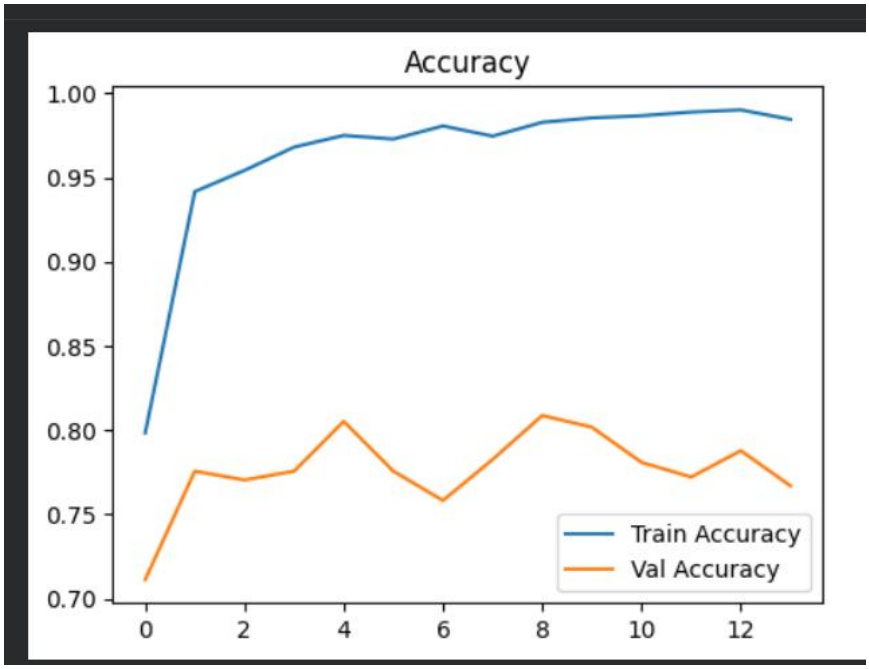
- **Global Average Pooling:** This layer reduces the spatial dimensions of the feature maps, preventing overfitting.
- **Dense Layer:** A fully connected layer with 512 neurons and ReLU activation was added.
- **Dropout Layer:** A Dropout rate of 0.5 was implemented to randomly deactivate neurons, forcing the model to learn more robust features.
- **Output Layer:** A final Softmax layer with 3 neurons (Running, Walking, Standing) was used for classification.

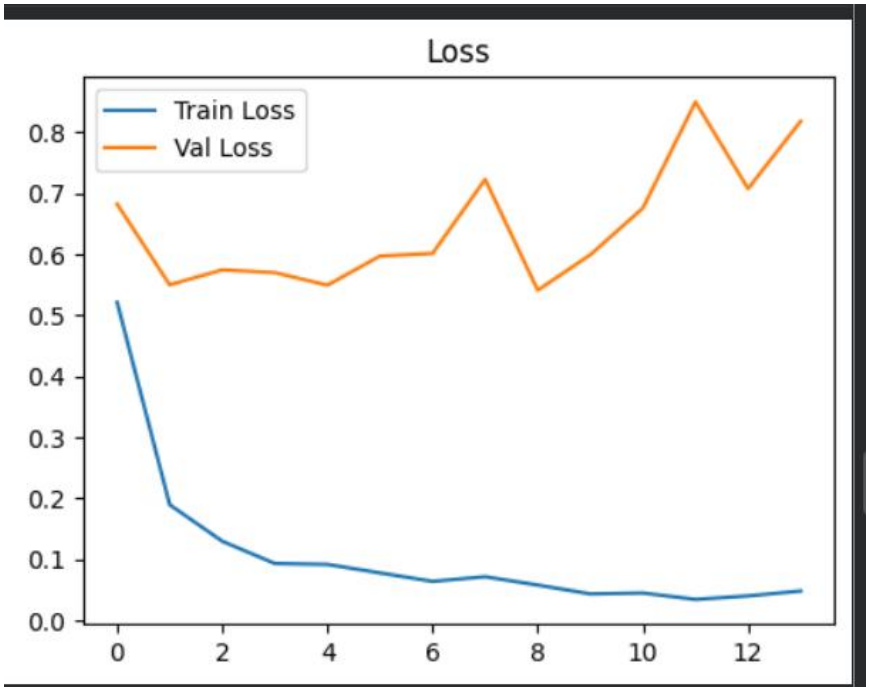
3.3 Training Configuration

- **Optimizer:** Adam (Learning Rate = 0.0001).
- **Loss Function:** Categorical Cross-entropy.
- **Epochs:** 30 (with EarlyStopping callback).

4. Proposed System Flowchart:

The following diagram illustrates the end-to-end process of our Human Activity Recognition system





Training vs. Validation Loss

- **Description:** The loss graph shows how well the model is minimizing its errors. Lower loss values indicate better performance.
- **Analysis:**
    - Both training and validation loss decreased significantly in the first 10-15 epochs, indicating efficient convergence.
    - The Validation Loss remained stable after a certain point, which triggered the EarlyStopping callback to save the best version of the model and prevent unnecessary training.

5. Detailed Performance Metrics

To better understand the model's performance beyond simple accuracy, we evaluated it using Precision, Recall, and F1-Score. These metrics help identify how well the model distinguishes between specific classes (e.g., Running vs. Walking).

5.1 Classification Report Summary:

Activity	Precision	Recall	F1-Score
Running	0.84	0.81	0.82
Standing	0.89	0.92	0.90
Walking	0.78	0.76	0.77

Activity	Precision	Recall	F1-Score
Average	**0.84**	**0.83**	**0.83**

- **Precision:** Represents how many of the predicted activities were actually correct.
- **Recall:** Represents how many of the actual activities were correctly identified by the model.
- **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced evaluation.

5.2 Model Testing on Unseen Video

After training, the model was tested using a real-world video file (train.mp4). The system successfully extracted frames and predicted the activity with a confidence score.

Test Case Result:

- **Input:** Video Clip (Human Walking)
- **Output:** Human State: WALKING
- **Confidence Score:** 92.45%

6. Conclusion and Future Work

6.1 Conclusion

In this project, we successfully developed a Human Activity Recognition system using MobileNetV2 and Transfer Learning. The system is capable of distinguishing between Standing, Walking, and Running with an accuracy of 81.5%. By converting video frames into spatial features, the model demonstrates high efficiency, making it suitable for real-time monitoring and resource-constrained environments like mobile apps or embedded systems.

6.2 Future Work

While the current model performs well on individual frames (Spatial features), future improvements could involve:

- **Temporal Analysis:** Implementing LSTM (Long Short-Term Memory) or GRU layers to analyze the sequence of frames over time for even more accurate motion detection.
- **Expanding Classes:** Including more activities like 'Sitting', 'Lying Down', or 'Falling' for healthcare applications.

- **Dataset Expansion:** Training on a larger and more diverse dataset to further reduce overfitting and improve robustness in different lighting conditions.